

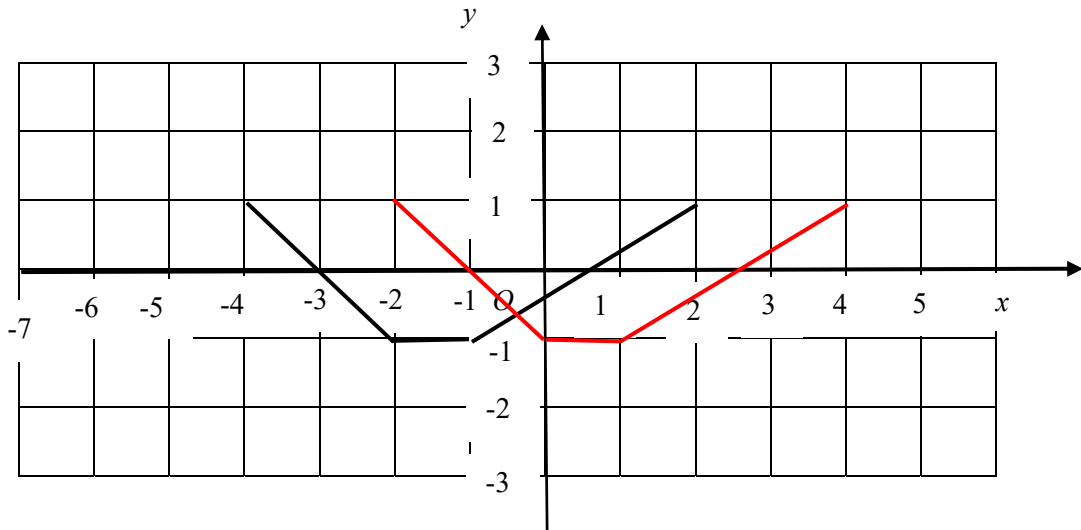
18	(a)	37, 143, 397, 503	M1	for any two correct angles within the ranges below or for a correct method to find a solution beyond 360, eg. “angle read from 0 to 360” + 360
			A1	for all 4 angles in the range, 35 to 40, 140 to 145, 395 to 400 and 500 to 505
	(b)	$y = -\sin x^\circ$	B1	for any acceptable equations, eg. $y = -\sin x^\circ$ or $y = \sin(-x^\circ)$ or $-y = \sin x^\circ$ or $y = \cos(x^\circ + 360n + 90)$ or for any positive integer n , $y = \sin(x^\circ - (2n - 1)180)$ or $y = \cos(x^\circ + 360n)$
	(c)	graph	C1	for correct graph shown translated 2 in the positive x -direction

Accept given as coordinates for M1 only

Quoted are just the more likely solutions but
check all attempts

Condone missing degrees sign

Question 18(c)



Documented scheme clarification

1MA1_NOV_2021_2H_Q18

Original source

scheme page 14, SHA256

585a47722d2f21280d403766b8edf4d7870d8afeea33cf868bfbb83c8e7199a8

Original wording

or $y = \cos(x \text{ (degrees)} + 360n)$

Correction / interpretation

Question 18(b): disregard the final cosine alternative. The reflected graph is $y = -\sin(x)$, with x in degrees. The earlier listed $y = \cos(x + 360n + 90)$ is equivalent for integer n .

Reason

For integer n , $\cos(x + 360n) = \cos(x)$, which is not $-\sin(x)$. At $x = 0$ the printed alternative gives $y = 1$, whereas the reflected sine graph gives $y = 0$. This clarification preserves the official source and its other valid alternatives.

Sol authority

dbe244ae769000276939275c1873b393c1746630e10e65487aa7f1cf259f21c1